

Introduction to Bayesian Reasoning

STAT 521 – Applied Probability and Statistical Inference

Brief History

- **Thomas Bayes (1701–1761)**: Presbyterian minister and mathematician in England.
- Worked privately on problems of inverse probability.
- His essay *“An Essay towards solving a Problem in the Doctrine of Chances”* was published posthumously in 1763 by Richard Price.
- Introduced the idea of updating probability beliefs with observed evidence.



Development and Revival of Bayesian Statistics

- **18th–19th century:** Pierre-Simon Laplace expanded Bayes' ideas.
- Most statistical inference was based on **inverse probability**, a precursor to modern Bayesian reasoning.
- **20th century:** Frequentist methods (Fisher, Neyman, Pearson) emerged and became standard.
- **1980s–1990s:** Computational revolution (MCMC, Gibbs sampling) made Bayesian inference practical at scale.
- **Today:** Bayesian methods are central to modern machine learning, hierarchical modeling, and decision-making.

What is Bayesian Statistics?

- A framework for statistical inference grounded in probability.
- Probability quantifies **degrees of belief** about unknown quantities.
- All uncertainty (e.g. parameters, hypotheses, predictions) is expressed probabilistically.
- Learning occurs by updating beliefs with data.

Frequentist vs Bayesian Thinking

Frequentist

- Probability = long-run frequency of events.
- Parameters are fixed but unknown.
- Data are random (from hypothetical repetitions).
- Inference via sampling distributions.

Bayesian

- Probability = degree of belief.
- Parameters are random variables.
- Data update beliefs.
- Inference via posterior distributions.

A One-Time Event: Will the Sun Come Out Tomorrow?

Frequentist definition: Probability = long-run frequency in repeated, identical experiments.

A One-Time Event: Will the Sun Come Out Tomorrow?

Frequentist definition: Probability = long-run frequency in repeated, identical experiments.

Problem:

- The event “sun rises tomorrow” is **not repeatable**.
- There is no sequence of tomorrows under identical conditions.
- Therefore, a Frequentist cannot formally assign a probability.

A One-Time Event: Will the Sun Come Out Tomorrow?

Frequentist definition: Probability = long-run frequency in repeated, identical experiments.

Problem:

- The event “sun rises tomorrow” is **not repeatable**.
- There is no sequence of tomorrows under identical conditions.
- Therefore, a Frequentist cannot formally assign a probability.

Implication:

- The event is either 0 or 1 (true or false), but unknown to us.
- Probability is a property of the sampling procedure, not the event.

Why “just use different days” does not work

- Frequentist probability requires an *identically repeatable experiment* with a genuine source of randomness.
- “Tomorrow” cannot be repeated under the same conditions—we cannot rerun the universe many times.

Why “just use different days” does not work

- Frequentist probability requires an *identically repeatable experiment* with a genuine source of randomness.
- “Tomorrow” cannot be repeated under the same conditions—we cannot rerun the universe many times.
- Different days are not identical trials (planetary, atmospheric, and astronomical conditions change).
- Solar mechanics are deterministic; uncertainty is in our knowledge, not in the data-generating mechanism.
- Therefore, the Frequentist framework provides no formal basis for assigning a probability to this one-time event.

Bayesian Perspective

Bayesian definition: Probability = degree of belief given current information.

A Bayesian can evaluate:

$$P(\text{Sun rises tomorrow} \mid \text{all astronomical knowledge})$$

Bayesian Perspective

Bayesian definition: Probability = degree of belief given current information.

A Bayesian can evaluate:

$$P(\text{Sun rises tomorrow} \mid \text{all astronomical knowledge})$$

Why this matters:

- We have overwhelming prior information:
 - Stable planetary orbits
 - Billions of consecutive sunrises
 - Stellar and orbital mechanics
- No evidence of imminent disruption.

Therefore: $P(\text{Sun rises tomorrow}) \approx 1$

Key difference:

- **Frequentist:** Cannot assign probability to a single, non-repeatable event.
- **Bayesian:** Naturally interprets probability as uncertainty about the world.

Summary

Key difference:

- **Frequentist:** Cannot assign probability to a single, non-repeatable event.
- **Bayesian:** Naturally interprets probability as uncertainty about the world.

Takeaway:

- Bayesian reasoning supports probability statements about *parameters*, *hypotheses*, and *one-time events*.
- This aligns closely with how people already think about uncertainty.

Another Example: Will It Rain Tomorrow?

- Weather report: “40% chance of rain tomorrow.”
- **Frequentist:** In infinitely many tomorrows with identical conditions, it rains 40% of the time.
- **Bayesian:** Given today's information, my degree of belief that it will rain is 40%.

Another Example: Will It Rain Tomorrow?

- Weather report: “40% chance of rain tomorrow.”
- **Frequentist:** In infinitely many tomorrows with identical conditions, it rains 40% of the time.
- **Bayesian:** Given today's information, my degree of belief that it will rain is 40%.

Takeaway

- Bayesians interpret probability subjectively (beliefs).
- Frequentists interpret probability via hypothetical repeated experiments.

Foundations of Probability

Joint Probability

$$P(A \cap B)$$

Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Product Rule

$$P(A \cap B) = P(A \mid B)P(B)$$

$$P(A \cap B) = P(B \mid A)P(A)$$

Bayes' Rule: Derivation

Start from the product rule:

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$$

Solve for the posterior probability:

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Where:

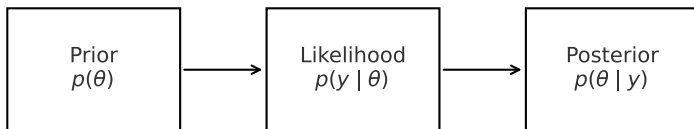
- $P(A) = \text{prior}$
- $P(B | A) = \text{likelihood}$
- $P(A | B) = \text{posterior}$
- $P(B) = \text{normalizing constant (marginal likelihood)}$

The Core Bayesian Identity

Bayesian Updating

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

- Prior: what we believe before seeing data.
- Likelihood: how the data are generated under the model.
- Posterior: updated beliefs after seeing data.



Prior, Likelihood, Posterior for a Parameter θ

Consider data y and an unknown parameter θ .

- **Prior** $p(\theta)$:
 - Encodes beliefs about plausible values of θ *before* seeing data.
 - Example: θ is a probability, so we might choose $\theta \sim \text{Beta}(\alpha, \beta)$.

Prior, Likelihood, Posterior for a Parameter θ

Consider data y and an unknown parameter θ .

- **Prior** $p(\theta)$:

- Encodes beliefs about plausible values of θ *before* seeing data.
- Example: θ is a probability, so we might choose $\theta \sim \text{Beta}(\alpha, \beta)$.

- **Likelihood** $p(y \mid \theta)$:

- Statistical model for how data are generated, given θ .
- Example: $y \mid \theta \sim \text{Binomial}(n, \theta)$.

Prior, Likelihood, Posterior for a Parameter θ

Consider data y and an unknown parameter θ .

- **Prior** $p(\theta)$:

- Encodes beliefs about plausible values of θ *before* seeing data.
- Example: θ is a probability, so we might choose $\theta \sim \text{Beta}(\alpha, \beta)$.

- **Likelihood** $p(y \mid \theta)$:

- Statistical model for how data are generated, given θ .
- Example: $y \mid \theta \sim \text{Binomial}(n, \theta)$.

- **Posterior** $p(\theta \mid y)$:

- Updated beliefs about θ after observing data.
- Given by Bayes' rule:

$$p(\theta \mid y) = \frac{p(y \mid \theta) p(\theta)}{p(y)} \propto p(y \mid \theta) p(\theta).$$

Contrast with a Frequentist Approach

Frequentist

- θ is a fixed but unknown constant.
- No prior distribution on θ .
- Likelihood $L(\theta; y) = p(y | \theta)$ is viewed as a function of θ .
- Estimate $\hat{\theta}$ via MLE; construct CIs and p -values using sampling distributions.
- No posterior $p(\theta | y)$.

Bayesian

- θ is a random variable.
- Specify a prior $p(\theta)$.
- Likelihood $p(y | \theta)$ combines with prior.
- Posterior
 $p(\theta | y) \propto p(y | \theta)p(\theta)$
summarizes uncertainty.
- Inference uses posterior means, medians, MAP, credible intervals, etc.

Confidence Interval vs Credible Interval: Big Picture

Frequentist Confidence Interval (CI)

- The parameter θ is *fixed but unknown*.
- The interval is random because it depends on the data.
- A 95% CI means: If we repeated the study infinitely many times, 95% of the confidence intervals would contain θ .
- **It does NOT mean** “there is a 95% probability that θ is in the interval.”

Confidence Interval vs Credible Interval: Big Picture

Frequentist Confidence Interval (CI)

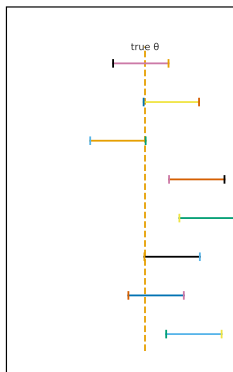
- The parameter θ is *fixed but unknown*.
- The interval is random because it depends on the data.
- A 95% CI means: If we repeated the study infinitely many times, 95% of the confidence intervals would contain θ .
- **It does NOT mean** “there is a 95% probability that θ is in the interval.”

Bayesian Credible Interval

- The parameter θ is treated as a *random variable*.
- We have a posterior distribution $p(\theta | y)$.
- A 95% credible interval means: Given the observed data, **there is a 95% probability that θ lies in the interval**.
- This expresses uncertainty about the value of θ directly.

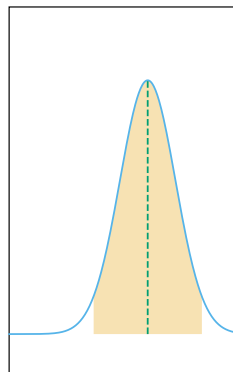
Visual Comparison

Frequentist confidence intervals



Each line = 95% CI from hypothetical repeated samples.

Bayesian credible interval



Posterior $p(\theta | y)$.
Shaded = 95% credible interval:

- Frequentist: Many hypothetical samples $\Rightarrow$ distribution of intervals.
- Bayesian: One posterior distribution $\Rightarrow$ probability about θ .

When Frequentist Approaches May Be Better

- **Very large datasets:** Priors matter less; MLE is fast and effective.
- **Simple models:** Closed-form estimators exist (e.g., classical regression).
- **Computational constraints:** MCMC may be too expensive.
- **Regulatory standards:** Some fields require frequentist tests and p -values.

When is Bayesian Statistics Preferred?

- **Small sample sizes:** Priors stabilize inference when data are limited.
- **Interpretability:** Credible intervals are often more intuitive than CIs.
- **Complex hierarchical models:** Natural and easier to specify.
- **Missing data or latent variables:** Posterior inference is straightforward.
- **Uncertainty quantification and propagation:** natural propagation through complex models.
- **Decision-making problems:** Posterior predictive distribution for predicting future outcomes.